ENGINEERING REFERENCE MONOGRAPH

Understanding Lithium Iron Phosphate (LFP) Technology

From Solid-State Crystal Kinetics to Non-Destructive Laboratory Diagnostics

Topic Domain:	Battery Energy Storage & EV Powertrains
Electrode Chemistry:	Olivine $\mathrm{LiFePO_4}$ Cathode / Mesocarbon Graphite Anode
Diagnostic Modalities:	UHPC, dQ/dV, DCIR Pulse Matrices, Hysteresis
Standards Alignment:	IATF 16949 • USABC • ISO 12405

Comprehensive Synthesis & Reference Documentation
Generated for Electrochemical Quality & Diagnostic Engineering

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Executive Summary: The Triad of Battery Diagnostics

Evaluating a battery without destroying it requires translating external terminal measurements (voltage, current, time, temperature) into internal microscopic states. In Lithium Iron Phosphate (LFP) systems, this is achieved through three complementary pillars:

1. **Ultra-High Precision Coulometry (UHPC):** Measures sub-microvolt parasitic side reactions to predict ten-year operational lifespan in weeks.
2. **Differential Capacity (dQ/dV):** Exploits the flat, motionless thermodynamic potential of the $\mathrm{LiFePO_4}$ cathode to expose the graphite anode's staging transitions as an internal "electrochemical MRI."
3. **Direct-Current Internal Resistance (DCIR):** Deconvolutes instantaneous ohmic tab resistance from interfacial charge-transfer and solid-state diffusion, dictating real-world EV acceleration and thermal cooling limits.

1. Introduction: The LFP Renaissance

For decades, the electric vehicle (EV) and energy storage systems (ESS) industries prioritized Nickel-Manganese-Cobalt (NMC) and Nickel-Cobalt-Aluminum (NCA) layered oxides due to their high gravimetric energy densities ($220 - 280\ \text{Wh/kg}$). However, the global market has experienced a massive strategic pivot back to Lithium Iron Phosphate (LiFePO_4):

- **Thermal Runaway Immunity:** The covalent $P-O$ bonds within the polyanionic phosphate framework do not release oxygen until temperatures exceed $500 - 600\ °C$, completely eliminating self-sustaining thermal runaway fires at standard operating temperatures.
- **Cycle Longevity:** Where high-nickel layered oxides degrade after $1,000 - 2,000$ full equivalent cycles due to anisotropic lattice contraction, LFP routinely achieves $3,000 - 10,000+$ cycles.
- **Geopolitical & Ethical Abundance:** LFP requires neither Cobalt (associated with severe supply-chain vulnerabilities and ethical concerns) nor Nickel (subject to extreme price volatility).

Electrochemical Parameter	Lithium Iron Phosphate (LFP)	High-Voltage NMC / LCO
Nominal Voltage (V_{cell})	3.20 V	3.70 V – 3.85 V
Voltage Window	2.50 V – 3.65 V	3.00 V – 4.35 V
Gravimetric Energy Density	140 – 180 Wh/kg	220 – 280 Wh/kg
Thermal Runaway Onset	$\approx 550°C$	$\approx 210°C$
Crystal Structure	Orthorhombic Olivine ($Pnma$)	Layered Rhombohedral ($R\bar{3}m$)
BMS State-of-Charge Difficulty	Extremely High (Flat Plateau)	Low (Monotonically Sloping)
Cycle Life (80% SOH)	3,000 – 10,000 Cycles	1,000 – 2,000 Cycles

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2. Solid-State Chemistry & The Olivine Framework

Lithium Iron Phosphate crystallizes in the **olivine structure** belonging to the orthorhombic space group $Pnma$.

2.1 The Polyanion Framework

The lattice framework is built from edge-sharing FeO_6 octahedra cross-linked by corner-sharing PO_4 tetrahedral units:

- Oxygen atoms form a slightly distorted, hexagonal close-packed arrangement.
- Phosphorus sits in tetrahedral interstitials, while Iron (Fe^{2+} in LiFePO_4, oxidising to Fe^{3+} in FePO_4) occupies octahedral sites.
- The strong sp^3 covalent hybridization of the $(PO_4)^{3-}$ polyanionic group dramatically lowers the energy of the $F\frac{e^{3+}}{F}e^{2+}$ redox couple via the inductive effect, stabilizing the redox potential at approximately $3.43\ V$ vs. $\frac{\text{Li}}{\text{Li}^+}$.

2.2 The 1D Diffusion Bottleneck

Unlike layered oxides where Li^+ ions diffuse in two-dimensional planes, the olivine lattice permits lithium transport strictly through **one-dimensional channels running along the** $[010]$ **crystallographic direction**:

$$D_{Li,[010]} \gg D_{Li,[100]} \approx D_{Li,[001]} \approx 0$$

> **The Iron Antisite Defect Mechanism**
>
> Because diffusion is strictly one-dimensional, any structural impurity sitting within the $[010]$ channel acts as a total roadblock. During high-temperature synthesis, iron atoms occasionally occupy lithium sites —known as **Fe-antisite defects** ($Fe_{Li}^{\cdot}$). A mere 1% concentration of iron antisite defects can reduce solid-state ionic conductivity by orders of magnitude by permanently blocking the channel.

2.3 The Nano-Carbon Solution

Pristine, bulk $\mathrm{LiFePO_4}$ is an electronic insulator with an electrical conductivity of $\sigma_e \approx 10^{-9}\ \mathrm{S/cm}$ and a low solid-state chemical diffusion coefficient ($D_{Li} \approx 10^{-14} - 10^{-16}\ \mathrm{cm^2/s}$). Two engineering breakthroughs enabled commercial viability:

1. **Particle Nano-Sizing (100–300 nm):** Decreasing grain radius (r) reduces the diffusion timescale exponentially, governed by the Einstein relation:

   $$\tau = \frac{r^2}{2D}$$

   Reducing particle size from $10\ \mu\mathrm{m}$ to $100\ \ \mathrm{nm}$ accelerates de-intercalation by four orders of magnitude (10^4).
2. **Pyrolytic Carbon Nano-Coating:** Applying an organic precursor (e.g., sucrose, glucose) pyrolyzed under inert atmosphere coats each nano-grain in a 2–5 nm graphitic carbon shell, raising effective conductivity to $\sigma_e > 10^{-2}\ \ \mathrm{S/cm}$.

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3. Thermodynamics: The Two-Phase Plateau & Gibbs Phase Rule

The electrochemical intercalation reaction of an LFP cathode is formulated as:

$$\mathrm{LiFePO_4} \underset{\text{discharge}}{\overset{\text{charge}}{\rightleftharpoons}} \mathrm{FePO_4} + Li^+ + e^- \quad \left(E^0 \approx 3.42 - 3.43\ V \ \text{ vs. } \frac{\mathrm{Li}}{\mathrm{Li^+}}\right)$$

3.1 The Miscibility Gap & The Gibbs Phase Rule

When lithium de-intercalates from $\mathrm{LiFePO_4}$, the system does not form a continuous solid solution. Instead, it splits into two distinct, coexisting thermodynamic phases across a wide miscibility gap:

- α-phase: Lithium-rich Triphylite ($Li_{1-\alpha}FePO_4$)
- β-phase: Lithium-poor Heterosite ($Li_\beta FePO_4$, where $\beta \approx 0$)

According to Gibbs' Phase Rule:

$$F = C - P + 2$$

Where:

- F is the number of intensive thermodynamic degrees of freedom.
- C is the number of chemical components ($C = 2$: $\mathrm{FePO_4}$ and Li).
- P is the number of coexisting phases in equilibrium ($P = 2$: Triphylite and Heterosite).

Under typical electrochemical laboratory testing at constant temperature (T) and pressure (P), the phase rule reduces to:

$$F = C - P = 2 - 2 = 0$$

Thermodynamic Consequence: Zero Degrees of Freedom

Because $F = 0$, the chemical potential of lithium (μ_{Li}) is strictly pinned and cannot vary with overall lithium concentration:

$$V_{\mathrm{OCV}} = -\frac{\mu_{Li}^{\mathrm{cathode}} - \mu_{Li}^{\mathrm{anode}}}{F_{\mathrm{Faraday}}} = \mathrm{constant}$$

This mathematical constraint is why the open-circuit voltage curve of $\mathrm{LiFePO_4}$ remains completely horizontal ($\approx 3.42\ V$ vs. $\frac{\mathrm{Li}}{\mathrm{Li^+}}$) across roughly 15% to 85% of its entire capacity.

3.2 The Latent Heat / Melting Ice Analogy

Engineers often struggle with the intuition that a battery is filling with capacity while its voltage does not budge. The thermodynamic equivalent is **melting ice at constant pressure**:

- When heat is poured into ice at $0°C$, the temperature stays at exactly $0°C$ until the last gram of ice melts. The energy is consumed entirely as latent heat of phase transformation.
- Similarly, while current (I) flows into an LFP cell on the flat plateau, Amp-hours are accumulating continuously:

$$Q(t) = \int_0^t I(t)\ dt$$

The electrical energy is consumed converting Triphylite crystal grains into Heterosite crystal grains without shifting the thermodynamic chemical potential.

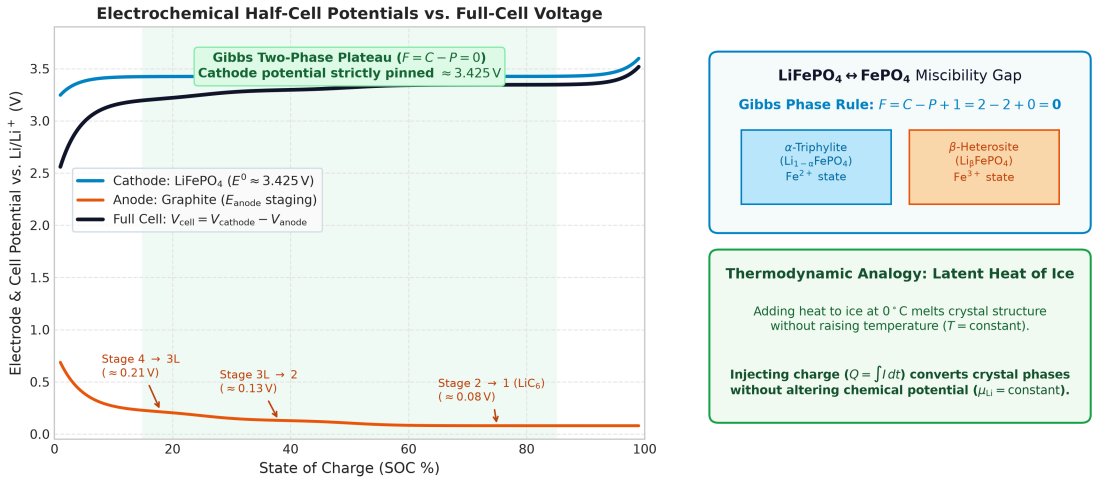


Figure 1: Electrochemical half-cell potential balance and the Gibbs phase rule equilibrium pinning the flat $\mathrm{LiFePO_4}$ cathode plateau.

3.3 The BMS Challenge: Why Voltmeters Fail

In an NMC or lead-acid battery, voltage correlates linearly with State-of-Charge (SOC). In LFP, the slope $d\frac{V}{d}Q \approx 0$:

- Between 20% and 80% SOC, terminal voltage varies by less than $10 - 20 \ \mathrm{mV}$.
- Normal sensor noise, temperature swings, and dynamic IR drops dwarf the open-circuit voltage variation.
- Consequently, Battery Management Systems (BMS) cannot rely on voltage lookups during operation. They must perform high-precision **Coulomb Counting** (integrating current over time) and calibrate only when the battery hits the upper saturation knee ($> 3.50\ V$) or lower depletion knee ($< 3.00\ V$).

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4. Differential Capacity Analysis (dQ/dV): The Non-Destructive MRI

Differential Capacity Analysis (dQ/dV) is the most powerful non-destructive analytical modality in battery physics. It converts flat, unreadable voltage plateaus into sharp, diagnostic mathematical peaks.

4.1 The Mathematical Proof: The Cathode as a Motionless Reference

The full-cell terminal voltage (V_{cell}) is the difference between cathode and anode potentials:

$$V_{\mathrm{cell}} = V_{\mathrm{cathode}} - V_{\mathrm{anode}}$$

Differentiating with respect to charge capacity (Q):

$$\frac{dV_{\text{cell}}}{dQ} = \frac{dV_{\text{cathode}}}{dQ} - \frac{dV_{\text{anode}}}{dQ}$$

Because the LiFePO_4 cathode operates within a two-phase equilibrium ($F = 0$), its derivative is essentially zero:

$$\frac{dV_{\text{cathode}}}{dQ} \approx 0$$

Substituting zero into the derivative:

$$\frac{dV_{\text{cell}}}{dQ} \approx -\frac{dV_{\text{anode}}}{dQ}$$

Inverting both sides:

$$\boldsymbol{\left(\frac{dQ}{dV_{\text{cell}}}\right) \approx -\left(\frac{dQ}{dV_{\text{anode}}}\right)}$$

> **The Built-In Reference Electrode**
>
> This mathematical identity proves that in an LFP/Graphite full cell, the cathode acts as a motionless, flat mirror. **Every single peak and valley observed on a full-cell LFP dQ/dV plot is a direct reflection of the graphite anode's internal staging reactions.**

4.2 Graphite Staging Deconvolution

Graphite (C_6) intercalates lithium through discrete, ordered stages rather than continuous absorption. Each stage produces a distinct plateau on the anode, mapped directly to full-cell voltage:

Graphite Stage	Intercalation Reaction	E_{anode} vs. $\frac{\text{Li}}{\text{Li}^+}$	Full Cell Peak ($3.43V - V_{\text{anode}}$)
Stage 4 $\rightarrow$ 3L	Dilute phase $\rightarrow LiC_{30}$	$\approx 0.20 - 0.22\ V$	Peak 1: $\approx 3.22 - 3.25\ V$
Stage 3L $\rightarrow$ 2	$LiC_{18} \rightarrow LiC_{12}$	$\approx 0.12 - 0.14\ V$	Peak 2: $\approx 3.30 - 3.32\ V$
Stage 2 $\rightarrow$ 1	$LiC_{12} \rightarrow LiC_6$ (Full)	$\approx 0.08 - 0.09\ V$	Peak 3: $\approx 3.34 - 3.36\ V$

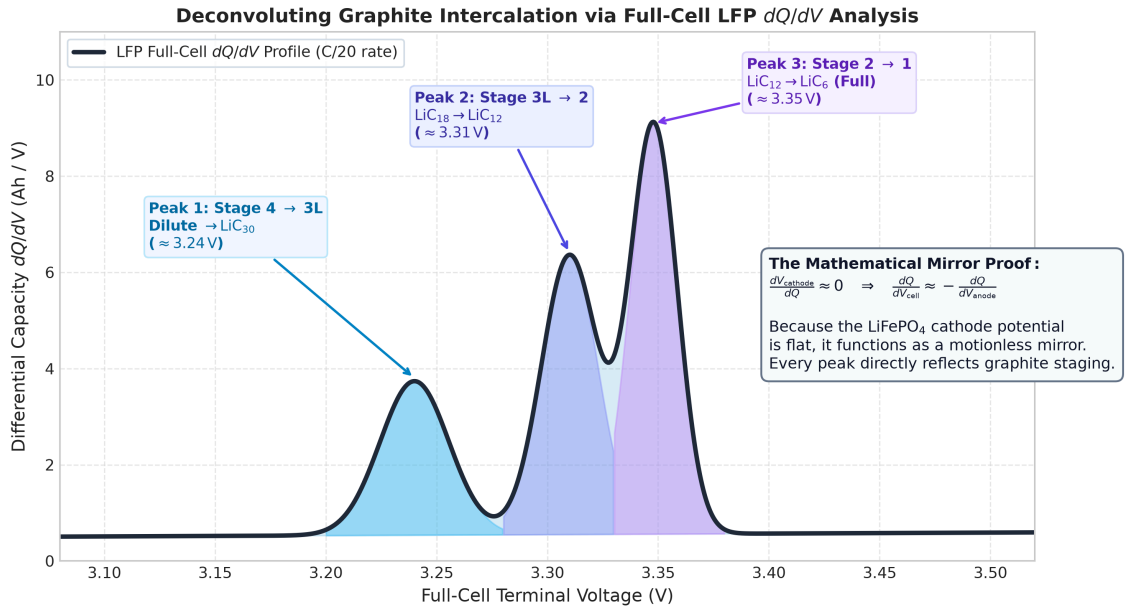


Figure 2: Differential capacity (dQ/dV) profile of an LFP full cell, exposing the graphite anode staging transitions through the motionless cathode mirror.

4.3 Deconvoluting the Three Fundamental Degradation Modes

Battery scientists use automated peak-fitting algorithms on dQ/dV curves to identify which degradation mode is destroying the battery:

1. **Loss of Lithium Inventory (LLI):** Active cyclable lithium ions get trapped permanently inside the growing Solid Electrolyte Interphase (SEI) layer on the graphite anode.
   - **Symptom:** The graphite peaks shift horizontally relative to the cathode capacity envelope, causing uniform peak shrinkage across all three peaks.
2. **Loss of Active Material on Anode ($\mathbf{LAM_{anode} = LAM_{NE}}$):** Graphite flakes exfoliate, crack, or become electrically isolated.
   - **Symptom:** The distance between Peak 1 and Peak 3 contracts; Stage 1 saturation (LiC_6) is truncated prematurely.
3. **Loss of Active Material on Cathode ($\mathbf{LAM_{cathode} = LAM_{PE}}$):** $\mathrm{LiFePO_4}$ particles crack or dissolve iron into the electrolyte.
   - **Symptom:** Total usable voltage window contracts; end-of-charge knee arrives early.

4.4 Electrochemical Hysteresis: Quantifying Round-Trip Energy Losses

When charge and discharge dQ/dV curves are plotted together, they do not overlap. The charge peak sits at a higher voltage than the discharge peak:

$$\Delta V_{\text{hysteresis}} = V_{\text{peak, charge}} - V_{\text{peak, discharge}}$$

- In pristine cells, $\Delta V_{\text{hysteresis}} \approx 20 - 30 \ \ \text{mV}$.
- This gap represents thermodynamic path-dependence (phase nucleation barrier) plus kinetic overpotential:

$$V_{\text{terminal}} = V_{\text{OCV}} \pm IR_0 \pm \eta_{\text{charge-transfer}} \pm \eta_{\text{diffusion}}$$

- **Thermal Waste Calculation:** Every cycle dissipates energy directly proportional to this voltage gap:

$$E_{\text{waste}} = \Delta V_{\text{hysteresis}} \times Q_{\text{cycle}}$$

Tracking hysteresis growth over cycling allows engineers to monitor cell impedance and size automotive cooling systems accurately.

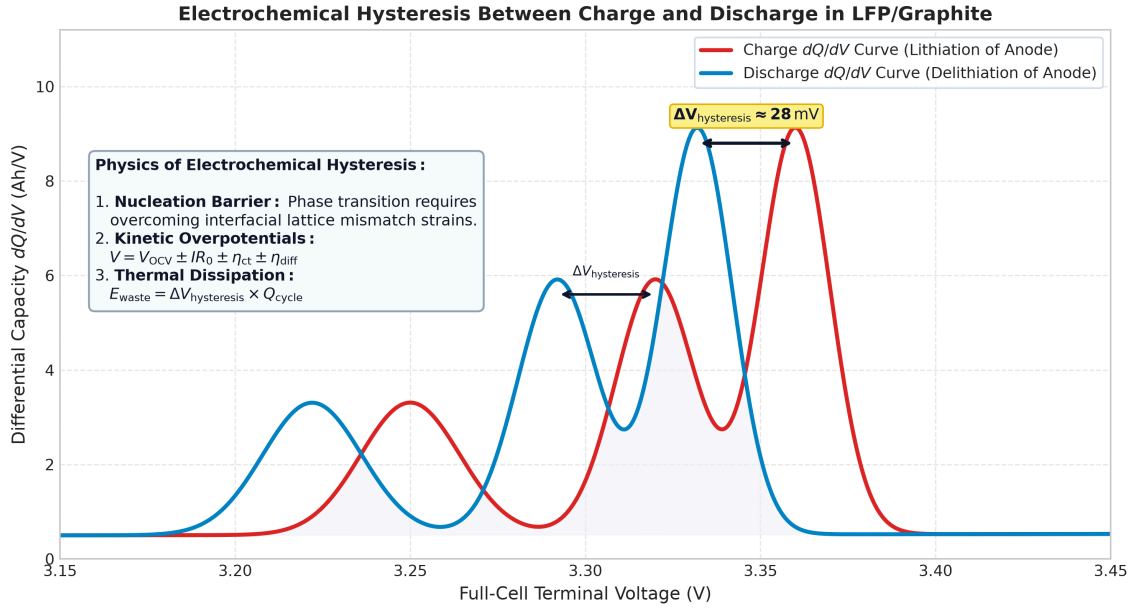


Figure 3: Electrochemical hysteresis gap ($\Delta V_{\text{hysteresis}} \approx 28 \text{ mV}$) between charge and discharge dQ/dV profiles, quantifying cycle thermal waste.

4.5 Catching Cold-Weather Metallic Lithium Plating

When an EV is fast-charged below $15°C$, slow solid-state diffusion causes the graphite surface overpotential to drop below the thermodynamic lithium plating threshold:

$$E_{\text{surface}} < 0.0 \text{ V} \quad \text{vs.} \quad \frac{\text{Li}}{\text{Li}^+}$$

Under this condition, lithium ions cannot intercalate into the carbon sheets; they precipitate directly as **metallic zero-valent lithium dendrites (Li^0)**.

- **The dQ/dV Detection Mechanism:** During subsequent discharge, metallic lithium strips off the graphite surface before intercalated lithium de-intercalates, creating a **distinct rogue mini-peak or shoulder** immediately at the onset of discharge ($\approx 3.40V$).
- This allows laboratory testing to establish exact safe charging envelopes without destructive cell teardowns.

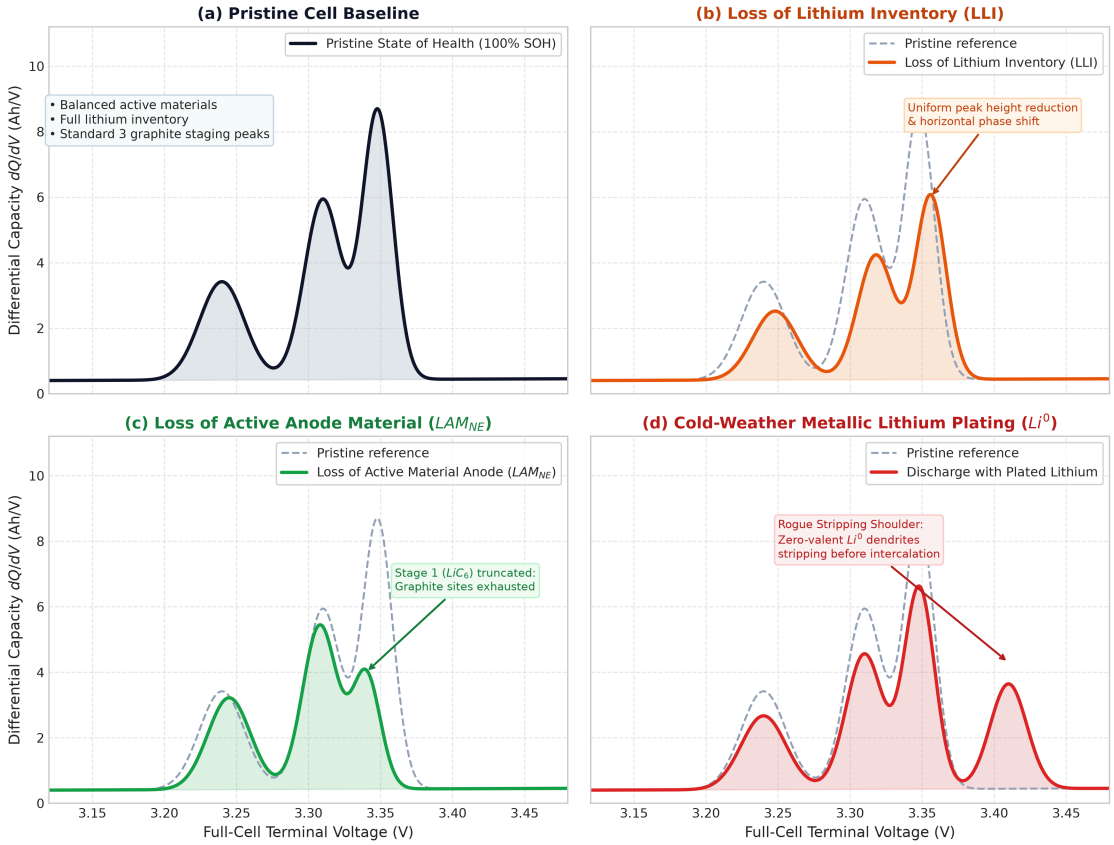


Figure 4: Deconvolution of battery degradation modes using differential capacity: (a) Pristine baseline, (b) Loss of Lithium Inventory (LLI), (c) Loss of Active Anode Material ($\mathrm{LAM_{NE}}$), and (d) Cold-weather metallic lithium plating.

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5. Direct-Current Internal Resistance (DCIR) & Pulse Polarization

While dQ/dV reveals chemical phase distribution, Direct-Current Internal Resistance (DCIR) measures high-rate power delivery and thermal safety.

5.1 Mathematical Extraction of Resistance

During a standard Hybrid Pulse Power Characterization (HPPC) step, a constant current pulse (I_{pulse}, e.g., $\pm 20\ A$) is injected for a fixed duration ($t = 10 - 20\ s$):

$$R_{\text{DCIR}} = \frac{|V_{\text{pre-pulse}} - V_{\text{end-pulse}}|}{|I_{\text{pulse}}|}$$

```
Terminal
Voltage
 V_pre ──────┐
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  <── R_0 (Ohmic Drop: Current collectors, tabs, electrolyte liquid)
    <── R_ct (Charge-Transfer Activation across SEI)
      <── R_diff (Solid-State Mass Transport Diffusion)
          V_end (measured at t = 20s)
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The voltage relaxation separates into three physical mechanisms:

1. **Instantaneous Ohmic Resistance (R_0, timescale: < 1 ms):** Bulk resistance of metal current collector foils, ultrasonic tab welds, and ionic conductivity of the liquid solvent.
2. **Charge-Transfer Resistance (R_{ct}, timescale: 1 ms $- 1$ s):** Activation energy for Li^+ to shed its solvation sheath and traverse the SEI membrane.
3. **Concentration Polarization / Diffusion (R_{diff}, timescale: $1 - 30+$ s):** Concentration gradients forming across the porous electrode thickness and within solid spherical particles governed by Fick's Second Law:

$$\frac{\partial C}{\partial t} = D_{solid}\left(\frac{\partial^2 C}{\partial r^2} + \frac{2}{r}\frac{\partial C}{\partial r}\right)$$

5.2 The SOC Bathtub Curve

When DCIR is plotted across State-of-Charge (SOC), it exhibits an asymmetric "bathtub" profile:

- **Middle Region (20% – 85% SOC):** Resistance remains uniformly low and flat ($\approx 6 - 8\ m\Omega$). Lithium ion vacancies and solid-state transport are well-balanced.
- **Low-SOC Region (< 10% SOC / $< 3.00\ V$):** Resistance spikes sharply ($> 14 - 20\ m\Omega$). At this point, the cathode is nearly saturated with lithium ions, leaving few empty sites for incoming flux, and liquid electrolyte pores suffer salt depletion.
- **BMS Inverter Power Derating:** Because $\Delta V = I \times R_{DCIR}$, drawing high current when resistance has doubled causes terminal voltage to crater beneath the hardware cutoff ($2.50\ V$). The car's BMS uses this exact curve to derate motor torque at low battery percentage.

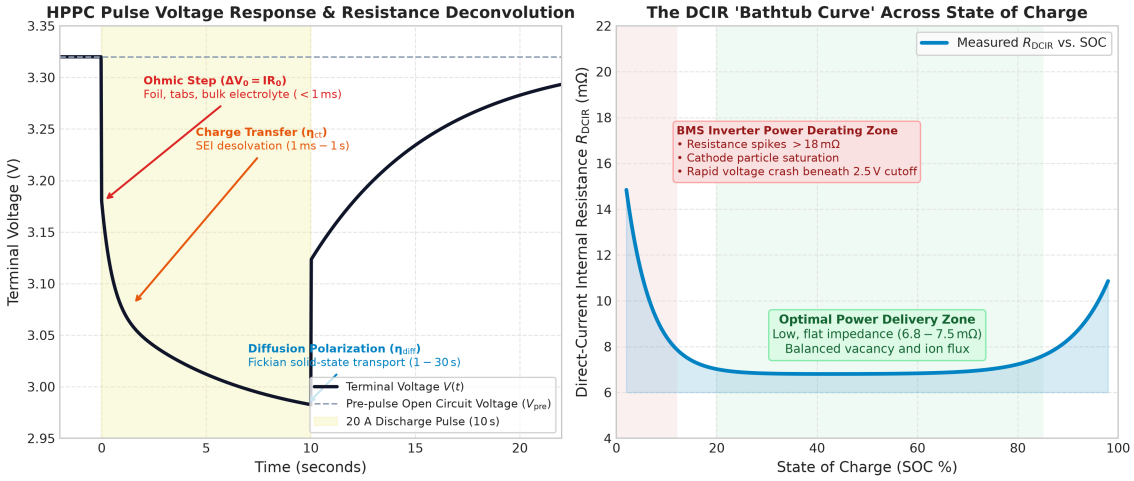


Figure 5: Direct-Current Internal Resistance (DCIR) dynamics: (Left) HPPC current pulse and voltage relaxation deconvolution; (Right) The DCIR "bathtub" curve across State-of-Charge highlighting low-SOC inverter power derating.

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6. Case Study: Resolving the Chemistry Mismatch

A critical real-world engineering lesson occurred during the analysis of cell `SLPBA842124HV-06`:

6.1 The Anomaly

The test dataset from SINTEF was labeled within an "LFP Platform" framework, but displayed characteristics that were physically impossible for Lithium Iron Phosphate:

- The cell operated between **3.00 V and 4.35 V** (resting open-circuit at 3.815 V).
- Charging true LFP to 4.35 V causes catastrophic electrolyte oxidation, severe outgassing, and thermal runaway.
- The dQ/dV curve exhibited **three prominent high-voltage peaks at 3.855 V, 3.99 V, and 4.10 V**, instead of the single flat LFP plateau.

6.2 The De-Identification

De-coding the cell model string resolved the paradox:

- **`SLPB`**: Kokam's trademark **Superior Lithium Polymer Battery** pouch architecture.
- **`A842124`**: Physical cell dimensions ($8.4 \text{ mm} \times 42 \text{ mm} \times 124 \text{ mm}$).
- **`HV`**: Explicitly designates **High Voltage** ($4.35\ V$ cutoff).
- **`Revolve_2024`**: Tested for Revolve NTNU (Formula Student electric racing team).

> **Engineering Takeaway: Trust the Thermodynamics**
>
> Software labels can be misapplied, but crystal thermodynamics cannot lie. The presence of three multi-phase de-intercalation peaks above 3.8 V definitively identified the cell as a High-Voltage Layered Oxide (NMC/LCO blend) rather than Lithium Iron Phosphate.

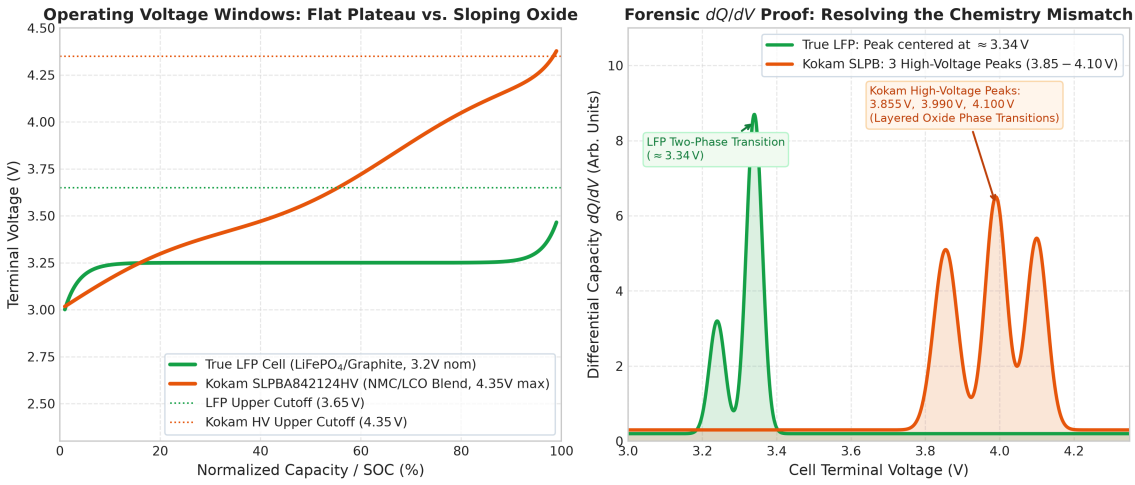


Figure 6: Forensic chemistry comparison: Flat LFP olivine plateau ($3.25\ V$ nom, $3.34\ V$ peak) versus sloping Kokam SLPB High-Voltage pouch cell ($4.35\ V$ max, 3 peaks at $3.855\ V$, $3.990\ V$, $4.100\ V$).

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7. Industrial Quality Engineering: Formation & SPC

Before an LFP cell enters an automotive battery pack, it must pass rigorous automated quality screening.

7.1 The Formation Paradox & Controlled SEI Synthesis

Freshly assembled pristine graphite has an operating potential ($< 0.1\ V$ vs. $\frac{\text{Li}}{\text{Li}^+}$) far below the thermodynamic reduction limit of organic alkyl carbonates ($0.8 - 1.0\ V$). Without a passivating layer, the electrolyte would reduce continuously until the battery died.

- **Uncontrolled Operation (Charging fresh cells at 1C):** Rapid reduction creates a brittle, porous, disorganized crust accompanied by violent gas generation (C_2H_4, CO, CO_2) that delaminates active layers.
- **Controlled Factory Formation (C/20 rates):** Slow, steady injection allows dense, compact inorganic inner films (LiF, Li_2CO_3) to form directly against the graphite face. This film blocks bulk solvent molecules while conducting Li^+ ions freely.

7.2 Micro-Short Screening via the K-Value

Microscopic burrs from current-collector slitting or separator pinholes create low-current leakage paths. After formation, cells sit in rest storage for 7 to 14 days, during which the cycler tracks the self-discharge rate (K-value):

$$K = \frac{\Delta V_{\text{OCV}}}{\Delta t} \quad [\text{mV / day}]$$

- Normal self-discharge for healthy LFP: $K < 0.20\ \ \text{mV/day}$.
- Quarantine threshold: $K \geq 0.50\ \ \text{mV/day}$.
- Any quarantined cell is removed from pack assembly immediately to eliminate spontaneous vehicle fires.

7.3 Statistical Process Control (SPC / IATF 16949)

In electric vehicles where hundreds of cells are wired in series, the pack is constrained by its weakest cell. Automakers apply Shewhart Statistical Process Control:

$$C_p = \frac{\text{USL} - \text{LSL}}{6\sigma}, \quad C_{pk} = \min\left(\frac{\text{USL} - \mu}{3\sigma}, \frac{\mu - \text{LSL}}{3\sigma}\right)$$

High-quality cell lines maintain $C_{pk} \geq 1.67$, ensuring that batch-wide capacity and DCIR variations are negligible.

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8. Appendix: Quick Reference & Mathematical Index

8.1 Key Electrochemical Formulas

Coulombic Efficiency (CE):

$$\mathrm{CE} = \frac{Q_{\mathrm{discharge}}}{Q_{\mathrm{charge}}} \times 100\%$$

Differential Capacity:

$$\frac{dQ}{dV} = \frac{I\ dt}{dV}$$

DC Internal Resistance (DCIR):

$$R_{\mathrm{DCIR}} = \frac{|\Delta V|}{|\Delta I|}$$

Joule Heat Dissipation:

$$P_{\mathrm{thermal}} = I^2 R_{\mathrm{DCIR}}$$

8.2 Degradation Fingerprint Reference Matrix

Observed Phenomenon	Physical Root Cause	Engineering Remedy
Uniform dQ/dV peak shrinkage	Loss of Lithium Inventory (LLI) via SEI growth	Optimize electrolyte additives (VC/FEC)
Widening $\Delta V_{\mathrm{hysteresis}}$ gap	Ohmic & charge-transfer impedance growth	Derate pulse C-rates; improve thermal cooling
Rogue shoulder on discharge dQ/dV	Metallic lithium plating on graphite surface	Increase low-temperature charge cutoffs
DCIR doubling ($> 100\%$ increase)	Mass-transport starvation or active particle loss	Trigger automotive End-of-Life (EOL) retirement
K-value $> 0.50\ \mathrm{mV/day}$	Internal micro-short circuit or separator defect	Scrap/quarantine cell before pack integration

— END OF TECHNICAL MONOGRAPH —
LFP Platform Research & Diagnostic Engineering Monograph